

Module 5

Two-Port Networks and Network Graphs

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Tie-set matrix

A tie – set is a set of branches contained in a loop such that each **loop contains one link and the rest are twigs.**

The set of branches forming the closed loop in which link or loop circulate is called a Tie-set.

From the knowledge of tie-sets (basic loops), the loop matrix can be formed.

In this matrix, the **loop orientation** is to be the same as corresponding **link direction**.

To find the tie-set matrix, first draw a graph and select a tree that connects all nodes without loops.

Then, identify the branches that are not in the tree (the links), as each link will form a fundamental loop with the tree branches.

Create a matrix with rows for the loops and columns for the branches, filling in +1 or -1 based on whether a branch's current aligns with the loop's direction, and 0 if it is not in the loop.

Tie-set matrix

Procedure to obtain the tie-set matrix:

Draw the graph and select a tree.

- i. Draw the network graph with nodes and branches.
- ii. Select a tree of branches which connects all nodes without forming any closed paths.
- iii. The branches in the tree are called “twigs” and the remaining are called “links”.

Identify fundamental loops:

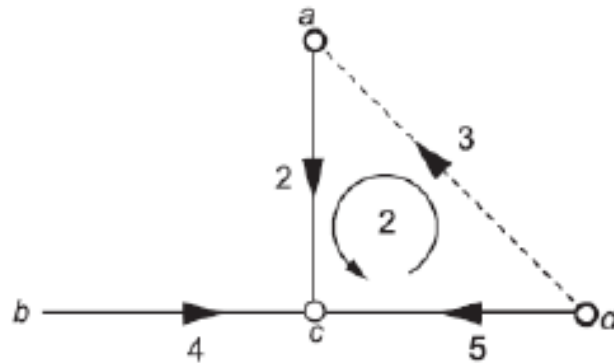
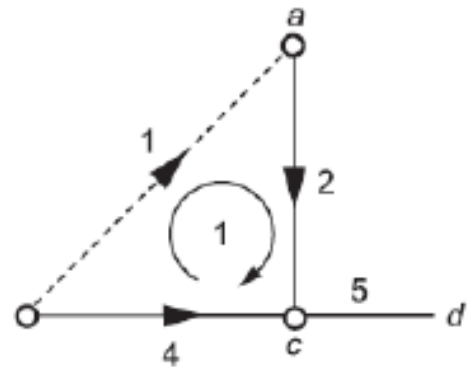
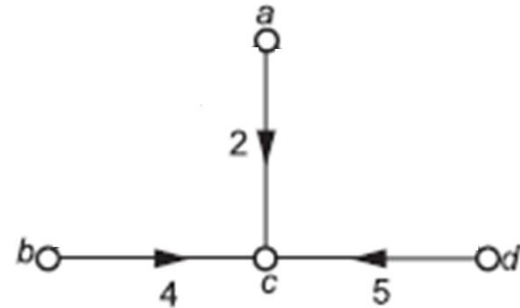
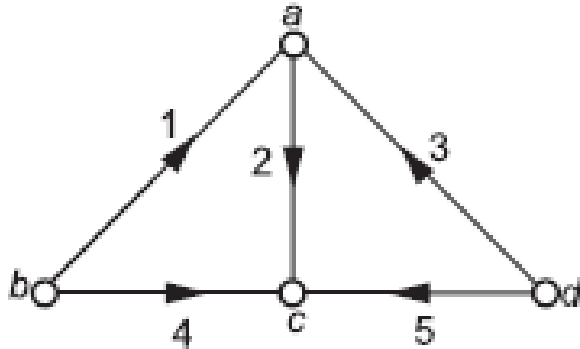
- i) Each link when added to twigs become a loop.
- ii) There will be a total of $B - N + 1$ fundamental loops, where B is the number of branches and N is the number of Nodes.

Form the tie-set matrix [B]:

- i) Create a matrix with the number of rows equal to the number of fundamental loops (the number of links) and the number of columns equal to the total number of branches.
- ii) Assign a direction to each fundamental loop in the same direction as its corresponding link.
- iii) For each loop (row): If a branch (column) is in the same direction as the loop, enter a +1, and -1 if the branch is in the opposite direction.

Tie-set matrix

1. Form the tie-set matrix for the graph shown considering the tree 2, 4, 5.



No. of link s (i)	1 (j)	2 (j)	3 (j)	4 (j)	5 (j)
1	1	1	0	-1	0
2	0	1	1	0	-1

$$[B] = \begin{bmatrix} 1 & 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \end{bmatrix}$$

Tie-set matrix

The **columns of [B]** gives the **branch currents** in terms of link currents.

Branch currents are represented as “j” and link currents are represented as “i”.

$$\left. \begin{aligned} j_1 &= i_1 \\ j_2 &= i_1 + i_2 \\ j_3 &= i_2 \\ j_4 &= -i_1 \\ j_5 &= -i_2 \end{aligned} \right\} \text{Eqn. 1}$$

The **rows of the [B]** give **KVL equations** in terms of the coefficient of the matrix.

$$\left. \begin{aligned} V_1 + V_2 - V_4 &= 0 \\ V_2 + V_3 - V_5 &= 0 \end{aligned} \right\} \text{Eqn. 2}$$

Tie-set matrix

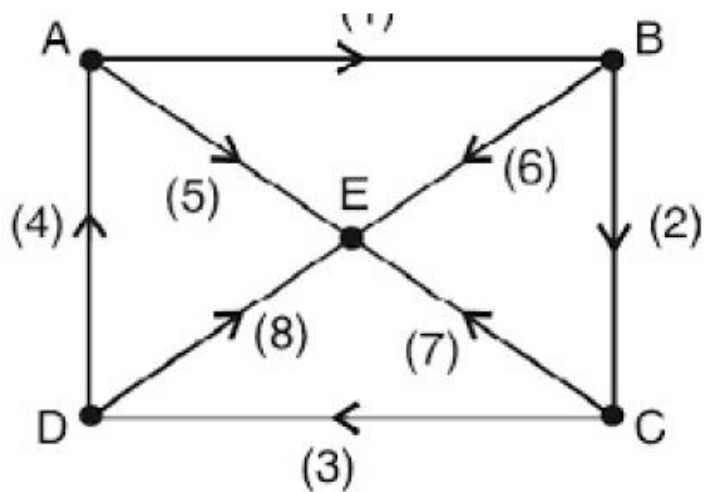
$$J_B = [B^T][I_L]$$

$$\begin{bmatrix} J1 \\ J2 \\ J3 \\ J4 \\ J5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} I1 \\ I2 \end{bmatrix}$$

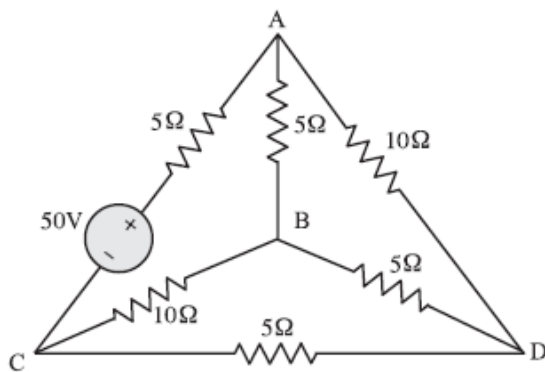
$$[B][V_B] = 0$$

$$\begin{bmatrix} 1 & 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} V1 \\ V2 \\ V3 \\ V4 \\ V5 \end{bmatrix} = 0$$

2. Form the tie-set matrix for the graph shown with respect to a tree.



3. Form the tie-set matrix for the network given below with respect to a tree.



Next class: Cut-set matrix